Restrictors

A door that slams is dangerous. How do we slow a piston down?



National 5 Engineering Science

2019 N5 Engineering Science: scan the code and try these.

Q10(a)

Page 16. Answer in your jotter.

Scan for the paper

Marking scheme

2023 N5 Engineering Science: scan the code and try these.

Q6(a)

Page 6. Answer in your jotter.

Scan for the paper

Marking scheme

By the end of this lesson… (p.13)

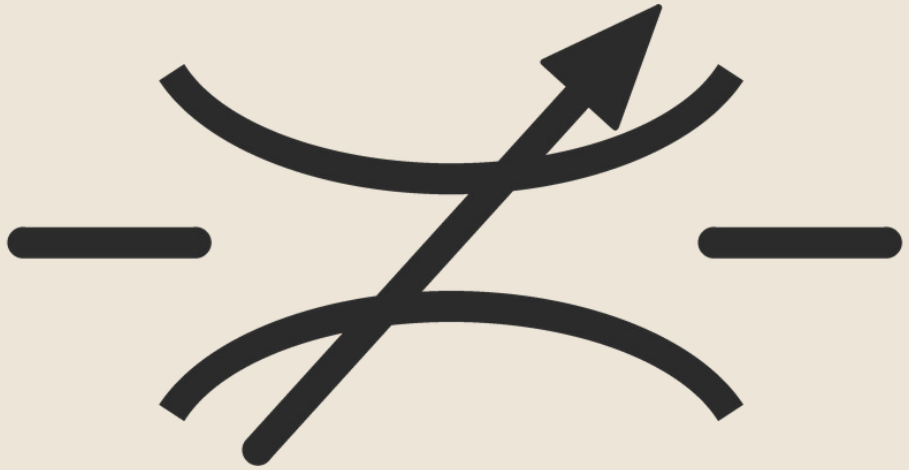


1. I can **identify** a restrictor and a uni-directional restrictor from their symbols.
2. I can **describe** how each one controls the speed of a piston.
3. I can **describe** where a uni-directional restrictor goes to slow only one stroke.
4. I can **draw** a uni-directional restrictor the right way round for the air flow.

The Restrictor (p.13)

A restrictor makes the path for the air **narrower**, so air flows more slowly in **both** directions.

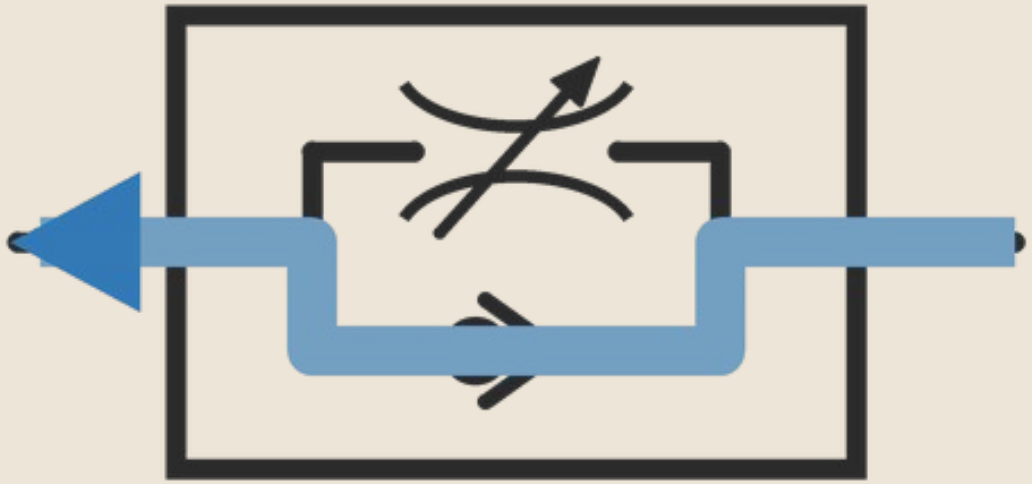
The arrow through it shows the restriction can be **adjusted**.



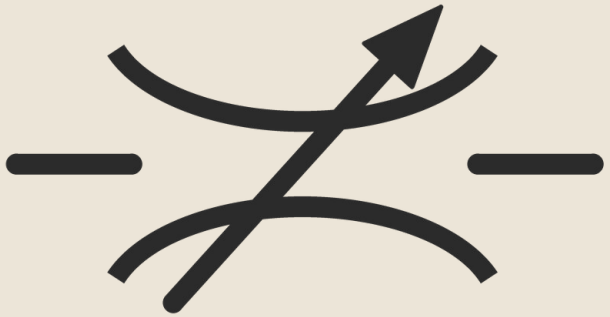
The Uni-Directional Restrictor (p.13)

A uni-directional restrictor slows the air in **one** direction only.

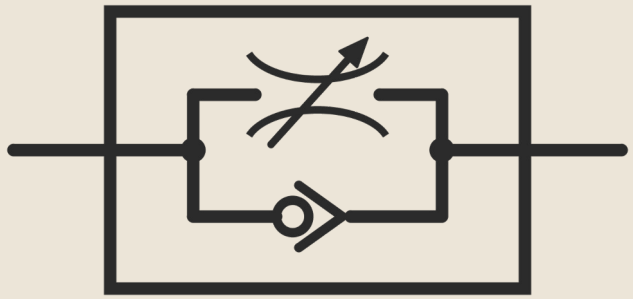
In the other direction the air flows freely through the **non-return** valve (the by-pass).



Show Me: Which One?



Restrictor: slows **both** ways



Uni-directional restrictor: slows **one** way

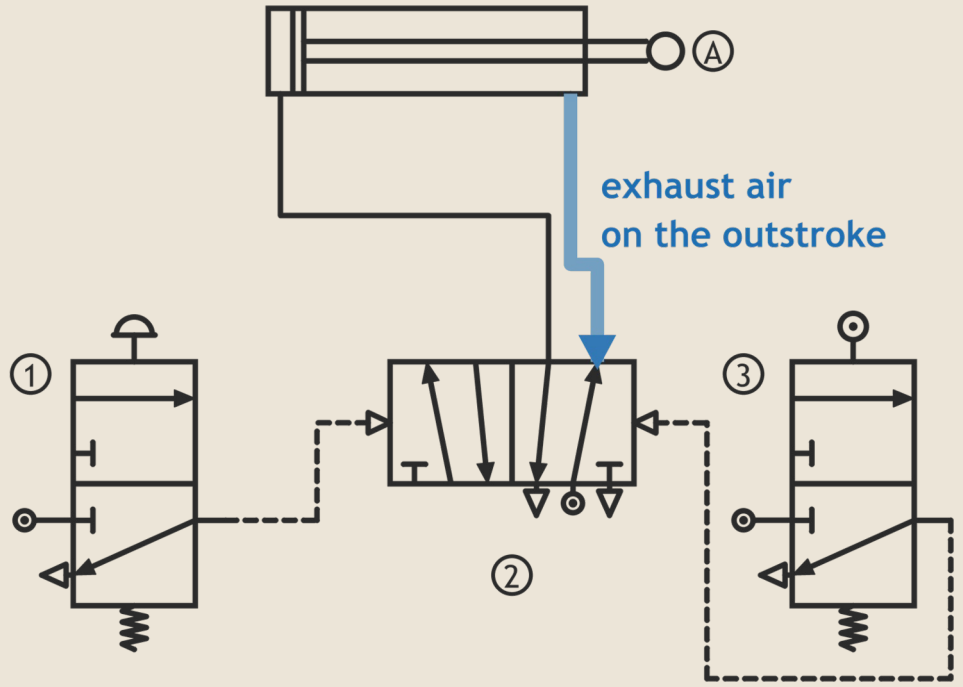


Non-return valve: lets air **one** way only

Where to Fit It (p.13)

To slow a piston **smoothly**, restrict the air **leaving** the cylinder (the exhaust air).

To slow the outstroke only, fit it in the pipe to the **instroking** side of the cylinder.



Words That Get the Marks (p.13)

Say how

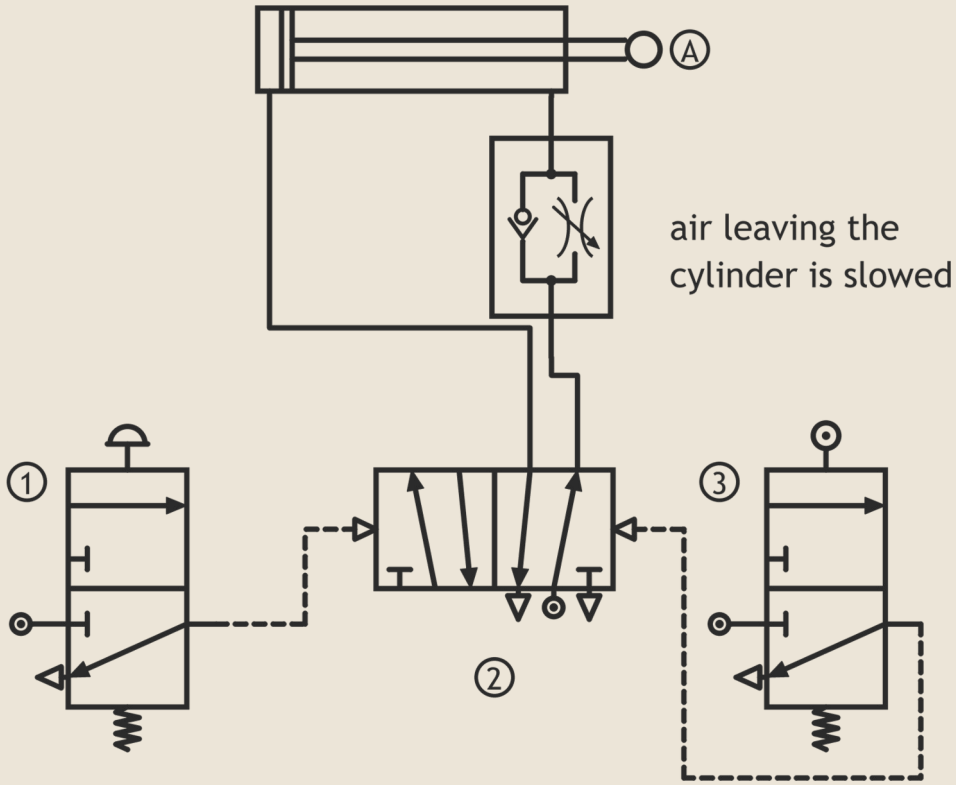
"restrict the **exhaust** air **leaving** the cylinder"

No mark

"UDR" on its own

Say where

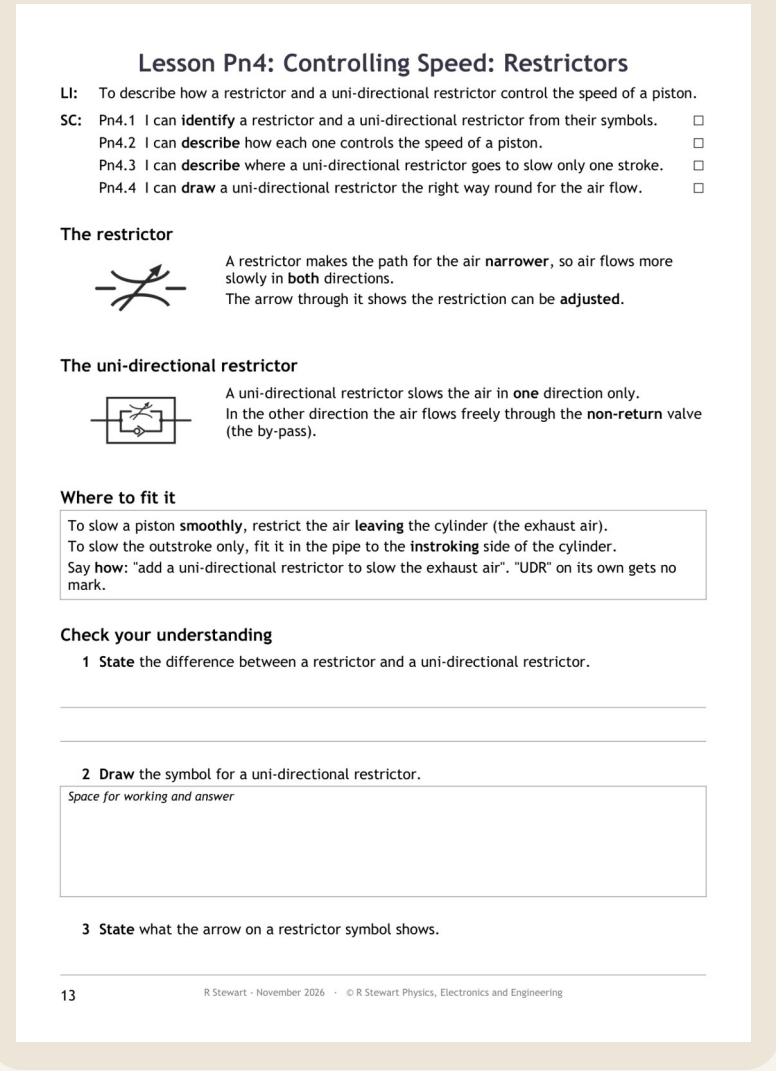
the pipe to the **instroking** side



Check Your Understanding (pp 13-14)

In your booklet

Answer Questions **1** to **5** on pages **13** and **14.**



Lesson Pn4: Controlling Speed: Restrictors

LI: To describe how a restrictor and a uni-directional restrictor control the speed of a piston.

SC: Pn4.1 I can **identify** a restrictor and a uni-directional restrictor from their symbols. ☐
Pn4.2 I can **describe** how each one controls the speed of a piston. ☐
Pn4.3 I can **describe** where a uni-directional restrictor goes to slow only one stroke. ☐
Pn4.4 I can **draw** a uni-directional restrictor the right way round for the air flow. ☐

The restrictor

A restrictor makes the path for the air **narrower**, so air flows more slowly in **both** directions.
The arrow through it shows the restriction can be **adjusted**.

The uni-directional restrictor

A uni-directional restrictor slows the air in **one** direction only.
In the other direction the air flows freely through the **non-return** valve (the by-pass).

Where to fit it

To slow a piston **smoothly**, restrict the air **leaving** the cylinder (the exhaust air).
To slow the outstroke only, fit it in the pipe to the **instroking** side of the cylinder.
Say **how**: "add a uni-directional restrictor to slow the exhaust air". "UDR" on its own gets no mark.

Check your understanding

1 **State** the difference between a restrictor and a uni-directional restrictor.

2 **Draw** the symbol for a uni-directional restrictor.

Space for working and answer

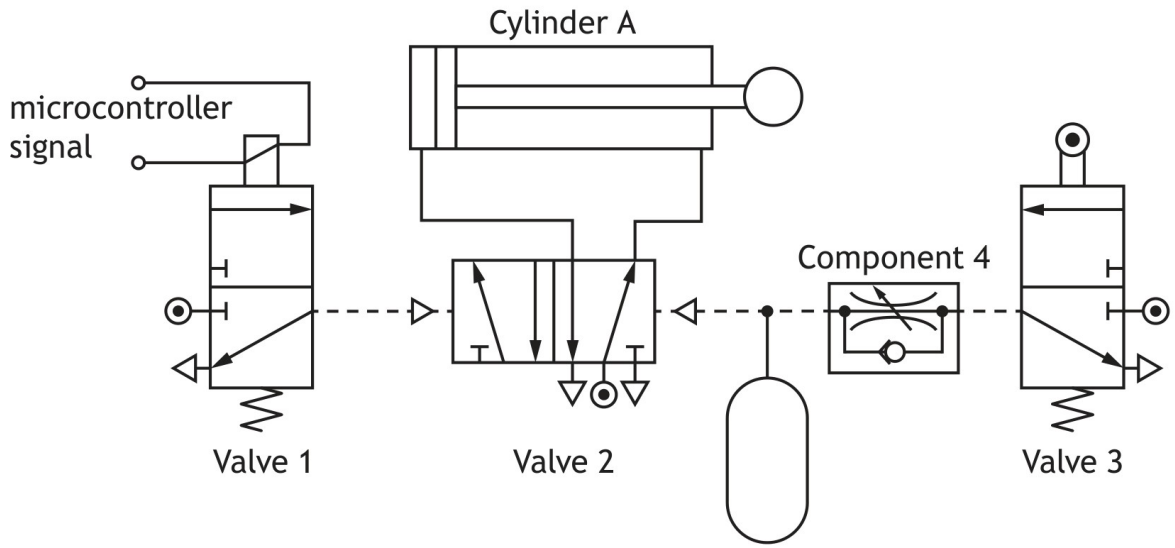
3 **State** what the arrow on a restrictor symbol shows.

Answers: Check Your Understanding (pp 13-14)

Question	Answer
1	Restrictor: **both** directions. Uni-directional: **one** only.
2	A restrictor with a **non-return valve** by-pass beside it.
3	The restriction can be **adjusted**.
4	The **exhaust** air: the air **leaving** the cylinder.
5	It does not say **what it does**: it **slows the exhaust air**.

Exam: 2016 Section 2 Question 12 (b)

Part of the pneumatic circuit used in the model is shown below.



Question 12 (b)

1 mark

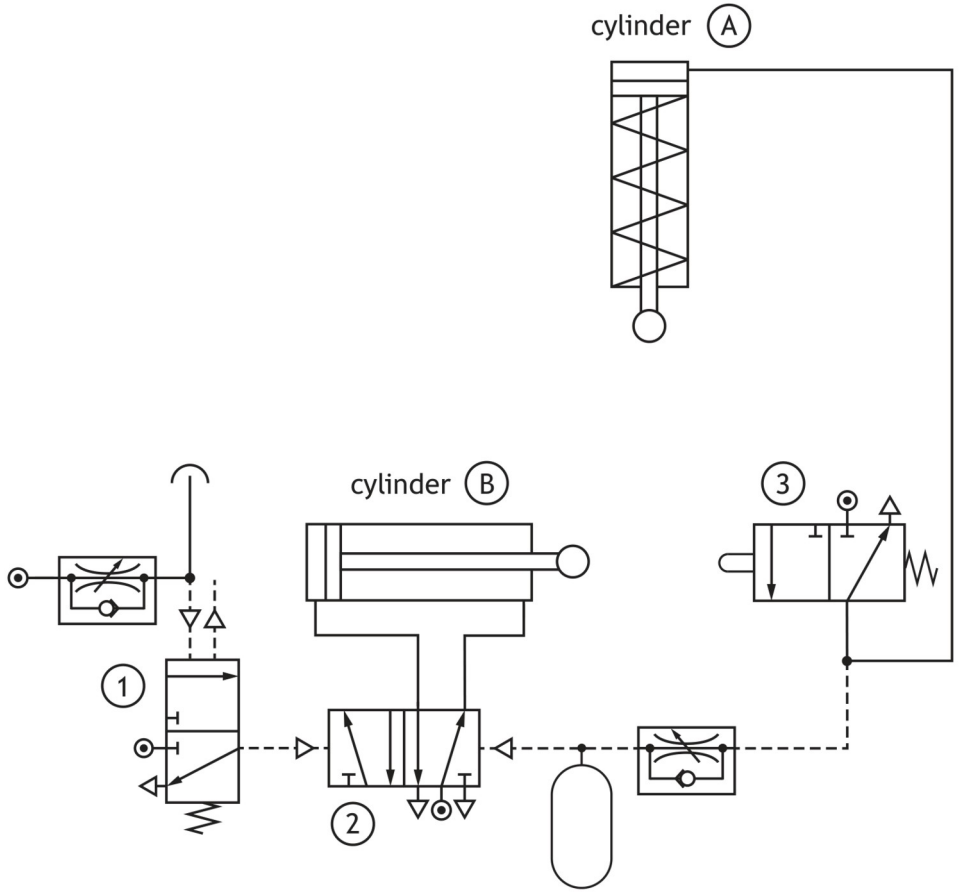
Page 14 of the paper.

The circuit is to be altered so that the piston **instrokes slowly**. **Indicate**, with an **X**, where a uni-directional restrictor should be connected.

Marking: 2016 Section 2 Question 12 (b)

Question			Expected Answer(s)	Max Mark	Additional Guidance
	(b)		Cross shown on left cylinder pipe.	1	Accept on the left hand exhaust port.

Exam: 2024 Section 2 Question 15 (c)



The piston in the double-acting cylinder is required to outstroke **slowly** and **smoothly**. The speed of the instroke should not be changed.

Question 15 (c)

2 marks

Page 30 of the paper.

Describe how this would be achieved.

Marking: 2024 Section 2 Question 15 (c)

Question	Expected response	Max mark	Additional guidance
(c)	Slow the air …on the 5/2 valve's exhaust port. Add a uni-directional restrictor …air in the instroking pipe of the double acting cylinder.	2	Descriptive response. 1 mark for restricting the air flow. 1 mark for exhaust air location. Accept responses shown on the circuit diagram. Do not accept 'UDR' on its own.

How It's Marked: Two Marks

1 mark

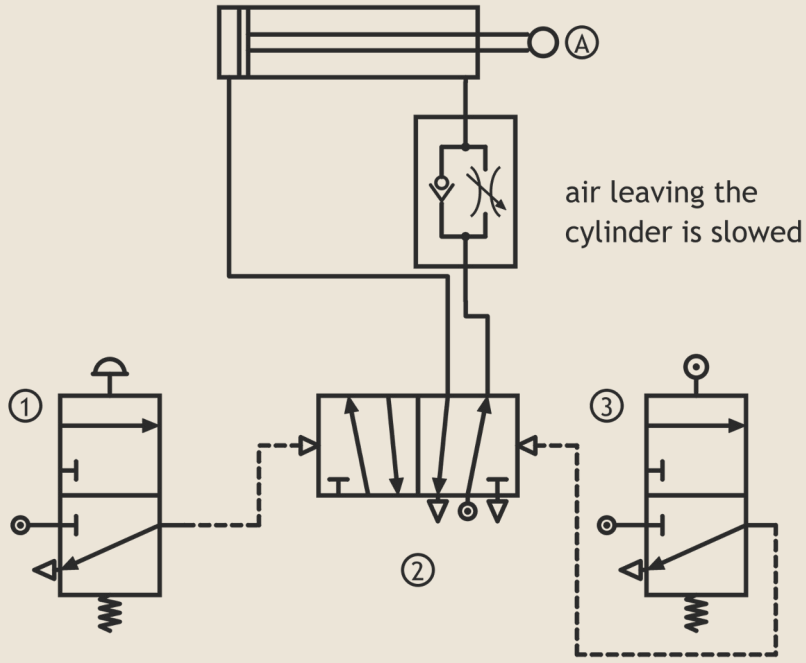
for **restricting** the air flow.

1 mark

for **exhaust** air location.

0 marks

"UDR" on its own.



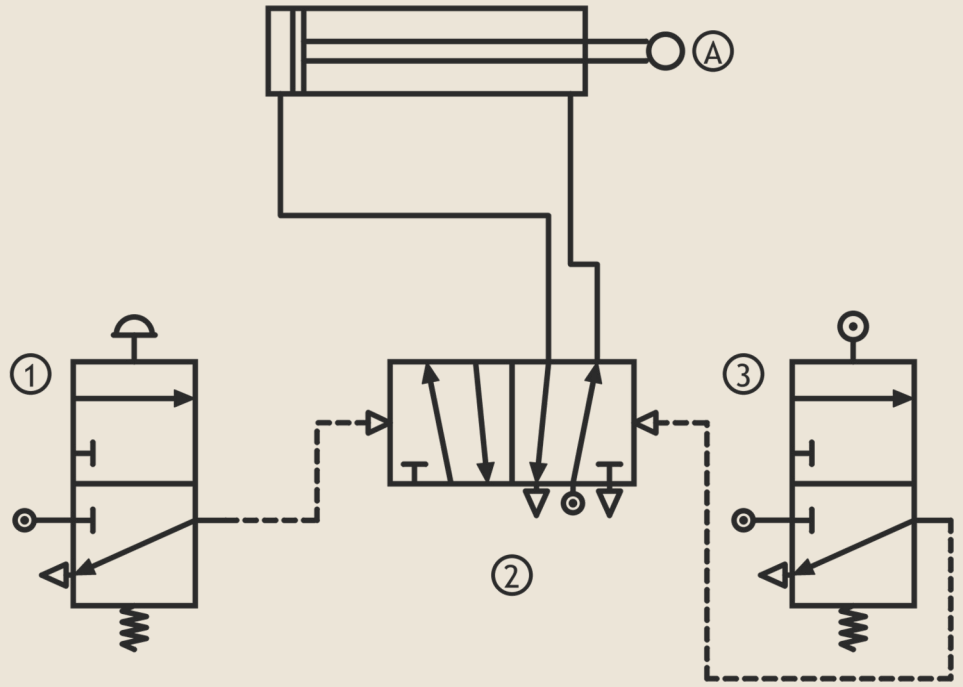
Build It: the Slow Door (p.14)

A piston opens a door. **Start** from the Pn3 circuit.

A Piston A **outstrokes** when valve 1 is actuated.

B Piston A outstrokes **slowly** and **smoothly**.

C The speed of the **instroke** is **not changed**.



Write a Test Plan (pp 14-15)

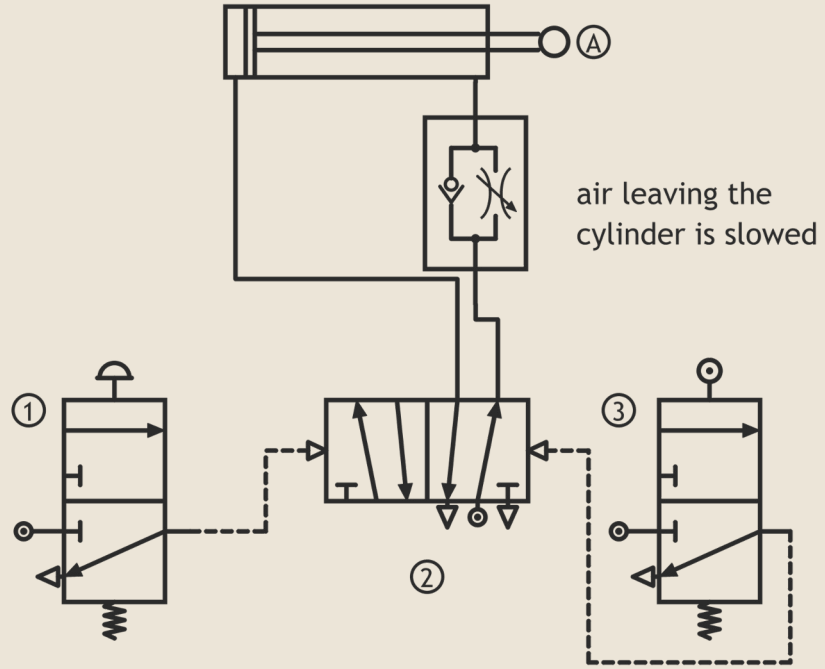


Planned test	**Expected result**	**Test result**
Actuate valve 1.	Piston A **outstrokes slowly** and **smoothly**.	
Watch piston A reach valve 3.	Piston A **instrokes** at **normal** speed.	
Repeat tests 1 and 2 three times.	The same **every time**.	

Simulate It (p.15)

In your booklet

1. **Open** NoPressureSim. **Build** your circuit, as on page 8.
2. **Predict** what happens. **Test** it.
3. Air off: **build** it on the kit. Air on: **test**.

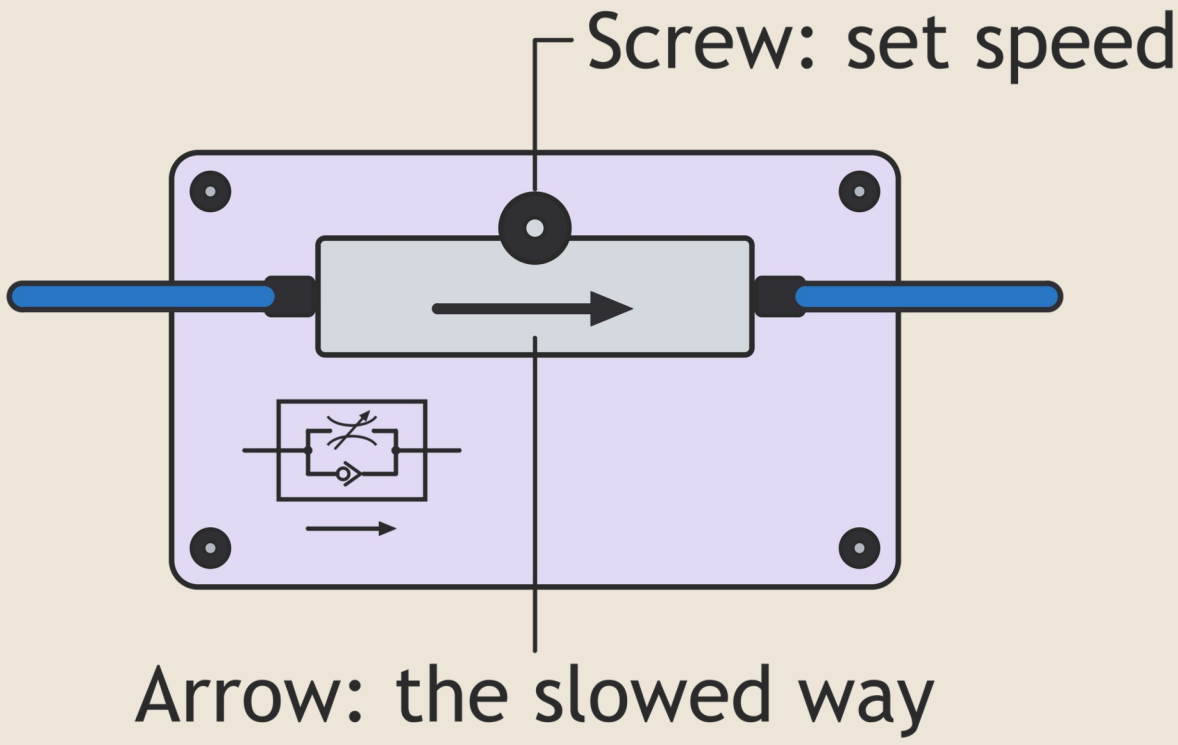


Scan me
NoPressureSim

What to Collect

Collect one set per pair.

1. 3/2 push-button valve
2. 5/2 pilot/pilot valve
3. 3/2 roller valve
4. Double-acting cylinder
5. Uni-directional restrictor
6. Eight blue tubes
7. Goggles each



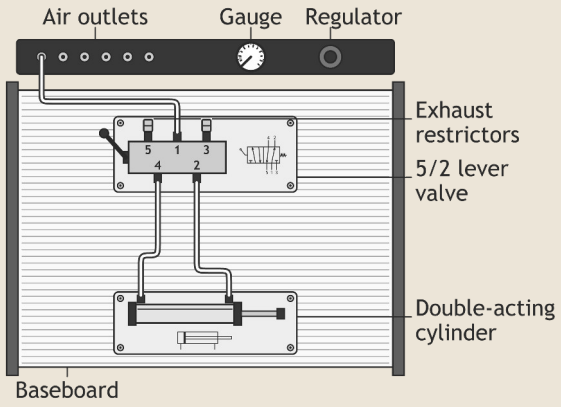
Build It on the Kit (p.15)

On the kit

Build it, air **off**.

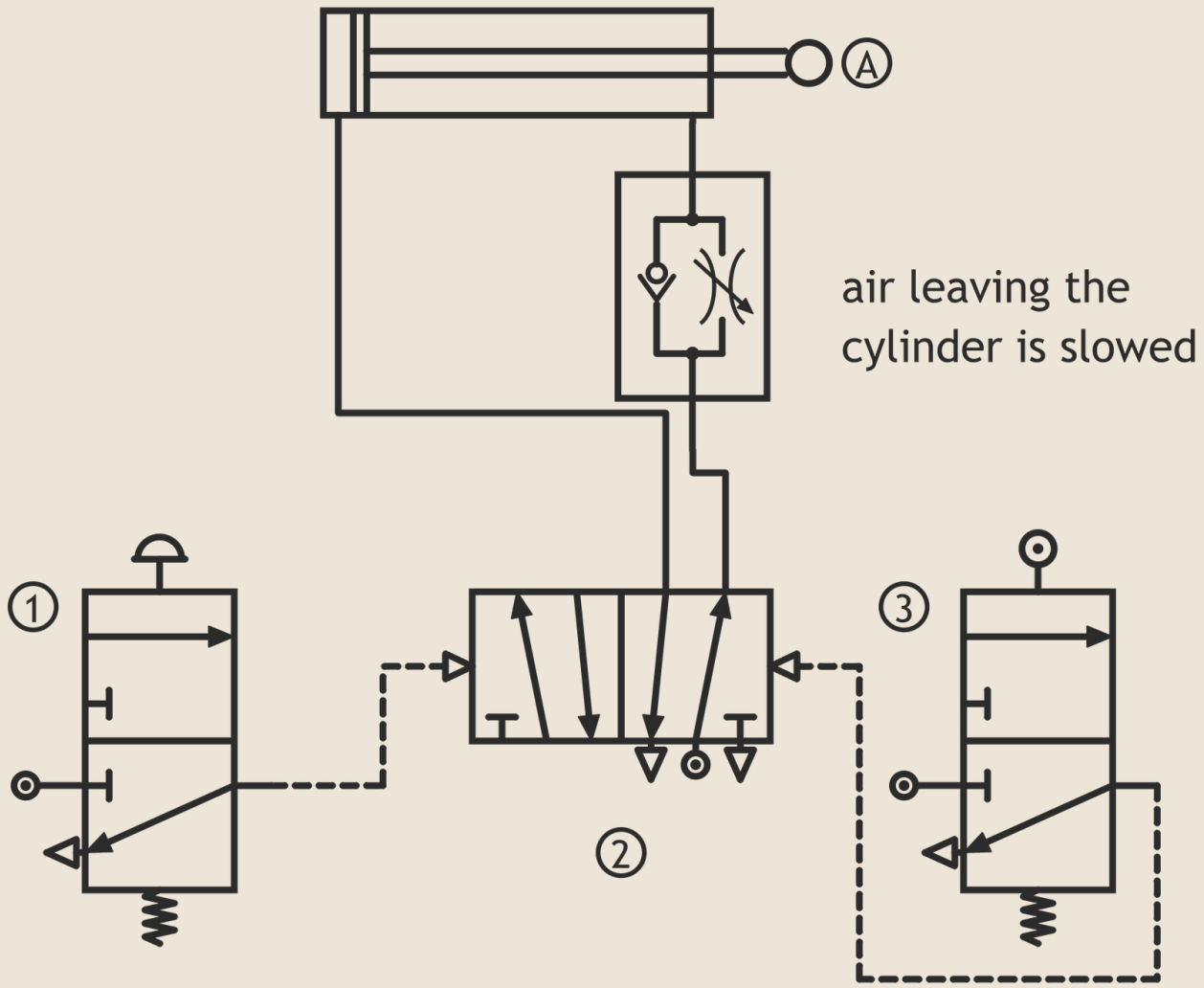
Restrictor in the pipe to the **instroking** side.

Air on. **Test** and **record**.



SAFETY 0 bar to build. 1 to 2 bar to test.

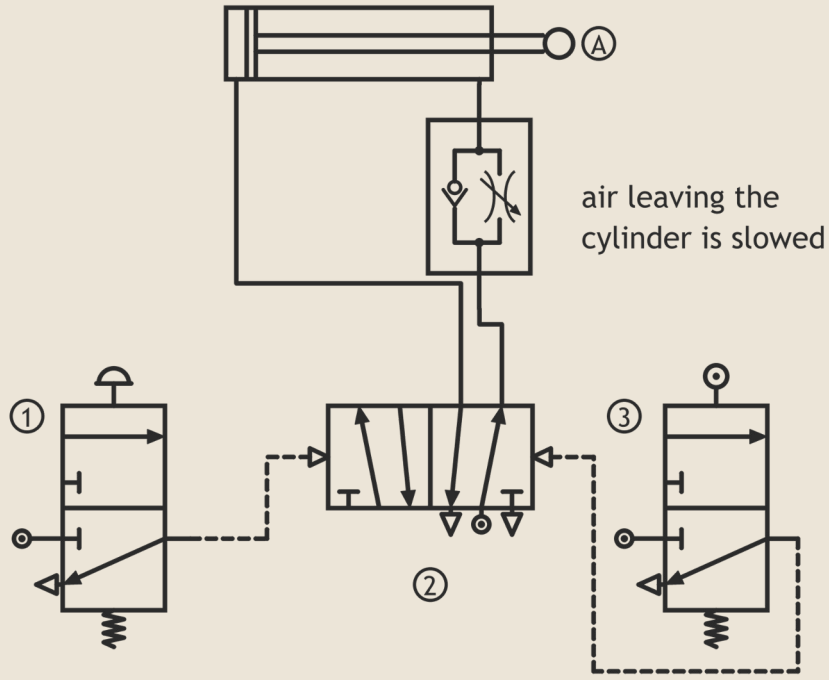
Build It: Bench Card 1



Evaluate (p.15)

In your booklet

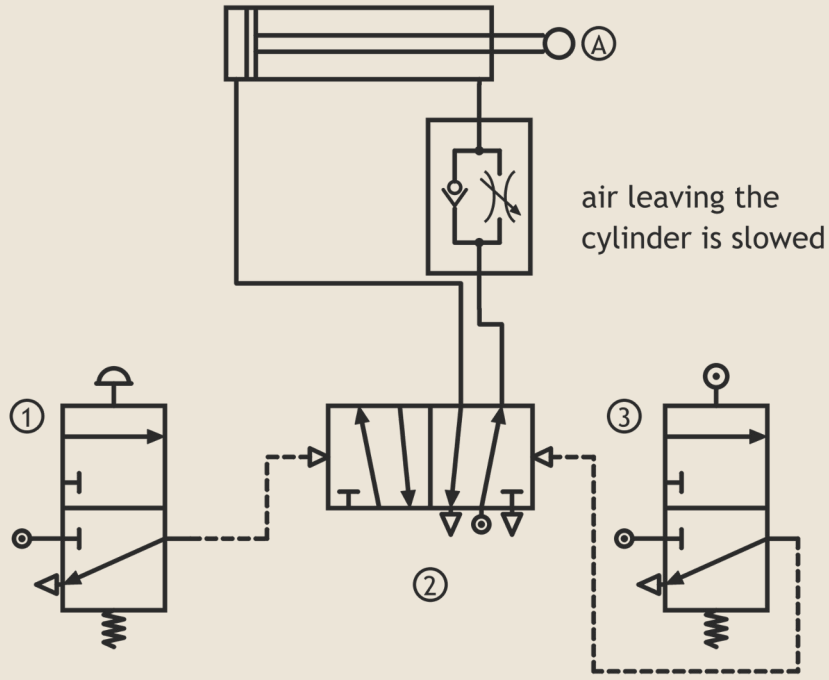
For each specification point, **state** if it was met.
Explain why, using your **test results**.
Say what you did **and** what the piston did.



Improve and Justify (p.15)

In your booklet

The door **closes too fast. Simulate** your fix only.
Describe a modification that would solve this.
Justify it: say **why** it solves the problem.



Answers: Evaluate (p.15)

Point	Met?	Explanation
A	Yes	**In test 1**, valve 1 was **actuated**, valve 2 **changed state**, **so** piston A **outstroked**.
B	Yes	**In test 1**, the restrictor **slowed the exhaust air**, **so** piston A outstroked **slowly** and **smoothly**.
C	Yes	**In test 2**, air passed **freely** through the **non-return** valve, **so** the instroke **did not change**.

Answers: Improve and Justify (p.15)

	Answer
Modification	A **second** uni-directional restrictor on the **other** pipe, slowing the **exhaust** air.
Justification	Piston A then **instrokes slowly** and **smoothly** too, **so** the door **closes safely**.

What Candidates Get Wrong

Say what it does

"UDR" on its own: 0 (2024 MI, 42 %)

Exhaust air

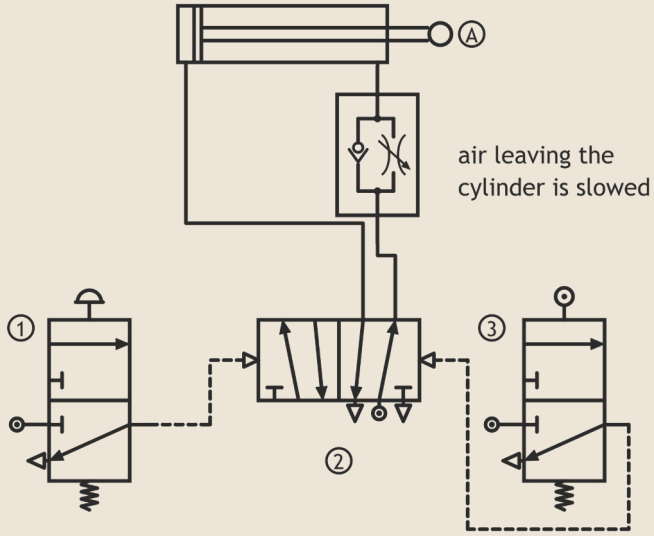
X on the **exhaust** pipe (2016, 2019 MIs)

Role and effect

explain what the restrictor does (2023)

Slowly and smoothly

many could not design it (2025)



Review Questions (p.14)

Lesson Pn4 (online)

Answer the Lesson **Pn4** questions in your **jotter**.
Scan the QR or go to **mrstewartphysics.co.uk/rq/pn4**.
Tap **Show answer** to **mark** each one.

Scan me

Look back — by the end of this lesson… (p.13)



1. I can **identify** a restrictor and a uni-directional restrictor from their symbols.
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3. I can **describe** where a uni-directional restrictor goes to slow only one stroke.
4. I can **draw** a uni-directional restrictor the right way round for the air flow.

Scan: page 6

Pneumatics

- symbols and operation of standard pneumatic components (restrictor, uni-directional restrictor, reservoir, 5/2 valve and actuators: diaphragm and solenoid)
- pneumatic time delay circuits
- calculating relationships between force, pressure and area in single-acting and double-acting cylinders
- controlling speed and force

Credits (1 of 2)

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Starter: 2019 N5 Engineering Science, © Qualifications Scotland (SQA). 2019 question paper and marking instructions; 2023 N5 Engineering Science, © Qualifications Scotland (SQA). 2023 question paper and marking instructions

Drawings: Pneumatic symbols and circuits drawn by R Stewart in the Qualifications Scotland paper style; CP2080 kit drawings by R Stewart.

Exam: 2016 Section 2 Question 12 (b): © Qualifications Scotland (SQA). 2016 National 5 Engineering Science Section 2, Q12

Marking: 2016 Section 2 Question 12 (b): © Qualifications Scotland (SQA). 2016 National 5 Engineering Science Section 2, Q12 marking instructions

Exam: 2024 Section 2 Question 15 (c): © Qualifications Scotland (SQA). 2024 National 5 Engineering Science Section 2, Q15

Credits (2 of 2)

Marking: 2024 Section 2 Question 15 (c): © Qualifications Scotland (SQA). 2024 National 5 Engineering Science Section 2, Q15 marking instructions

How It's Marked: © Qualifications Scotland (SQA). 2024 National 5 Engineering Science Section 2, Q15 marking instructions

Screenshots and page figures: Booklet 6, Pneumatics (N5 Engineering Science, R Stewart, November 2026): page images.

What Candidates Get Wrong: © Qualifications Scotland (SQA). National 5 Engineering Science course reports 2023 and 2025; 2024 S2 Q15(c), 2016 S2 Q12(b) and 2019 S2 Q10(c) marking instructions

Course specification: © Qualifications Scotland (SQA). N5 Engineering Science course specification v2.0, p.6

R Stewart - November 2026